



Internship Final Report

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Internship Duration	1 ST September to 30 TH September
Company Name	ShadowFox
Role	Data Analyst
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Project Title

Decoding Starbucks: An Exploratory Analysis of Calories, Sugars, and Nutritional Metrics

Objectives:

Comprehensive Nutritional Data Exploration:

To thoroughly analyze the Starbucks beverage menu dataset (starbucks_drinkMenu_expanded.xlsx), identifying underlying distributions, statistical variances, and key nutritional parameters— specifically focusing on calories, total sugars, and caffeine content across diverse drink categories.

Advanced Data Visualization & BI Dashboard Architecture:

To design and develop a dynamic, user-centric, and interactive **Power BI Dashboard** that transforms raw numerical data into clear visual formats, featuring custom canvas formatting, a cohesive color palette, and structured layout panels.

Interactive Filtering & Dynamic Analytics Implementation:

To implement multi-level slicers and category-based filtering mechanisms that allow users to dynamically explore specific beverage categories, preparation types, and individual drink items while maintaining real-time cross-filtering across all visual elements.

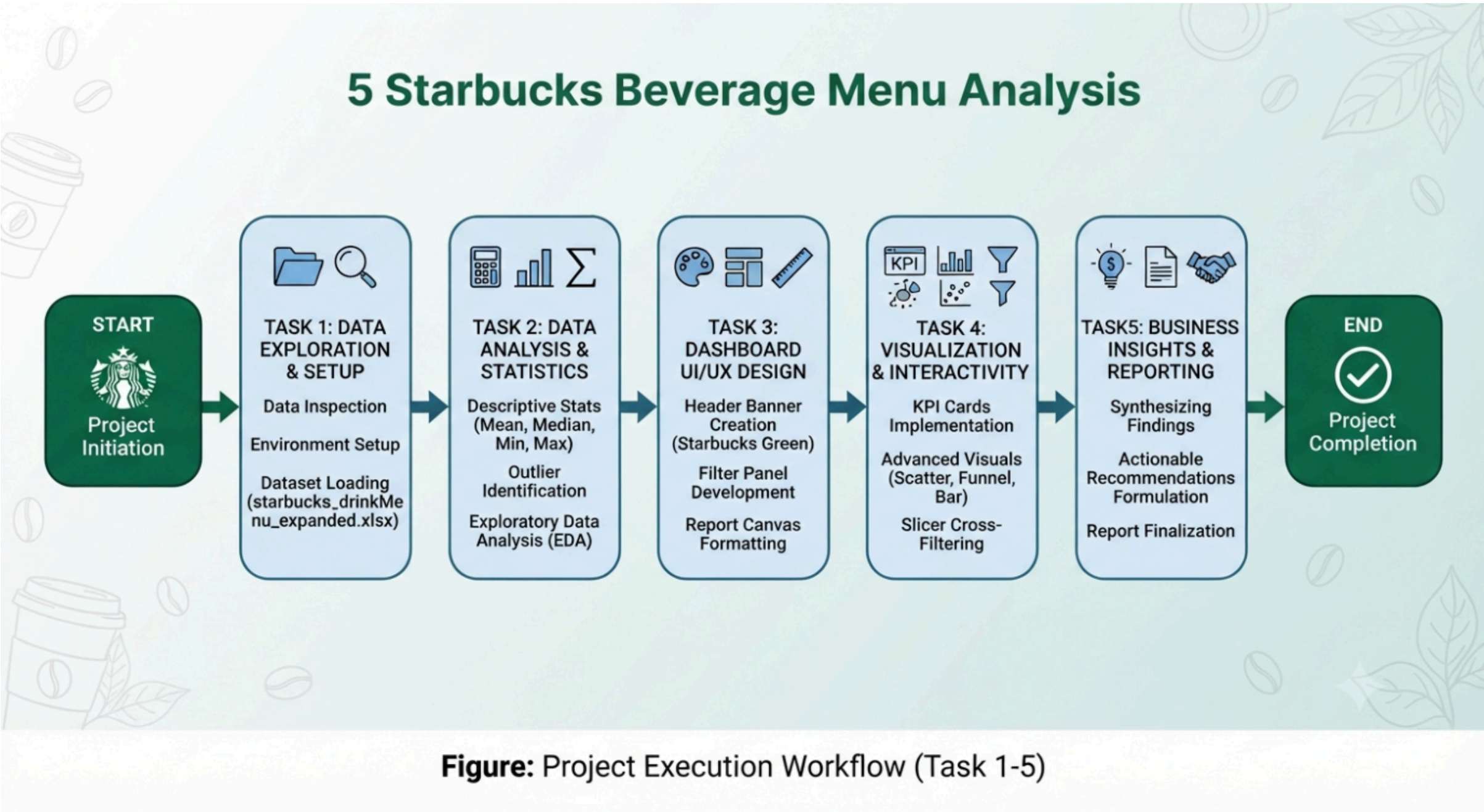
Correlation & Behavioral Pattern Identification:

To examine the statistical relationships between critical dietary attributes— such as evaluating how sugar content influences overall caloric load across handcrafted and specialty beverages using advanced scatter plots and distribution charts.

Actionable Business Insights & Health-Centric Recommendations Formulation:

To translate data-driven findings into strategic, value-driven business insights, offering actionable recommendations for the food and beverage industry to promote transparency, highlight low-calorie alternatives, and support informed, health-conscious consumer choices.

Flow Diagram:



Tasks and Responsibilities:

Task 1: Data Exploration

1.1 Dataset Overview & Source Identification

- The analysis is performed using the `starbucks_drinkMenu_expanded.xlsx` dataset, which captures comprehensive nutritional information and product characteristics for various beverages served at Starbucks.

1.2 Core Variables and Schema Analysis

- To understand the structure of the dataset, key columns and parameters were thoroughly examined:
 - **Beverage_category:** Categorizes drinks into broad groups (e.g., Coffee, Tea, Smoothies) to facilitate segmented analysis.
 - **Calories:** Quantifies the total energy derived from each beverage serving.
 - **Sugars (g):** Measures carbohydrate/sugar content, serving as a vital metric for sweetness and dietary impact.
 - **Total Fat (g):** Captures the fat profile, distinguishing dairy-heavy items from lighter bases.
 - **Caffeine (mg):** Tracks stimulant levels across different coffee and tea preparations.

1.3 Data Integrity & Preliminary Profiling

- Checked for null values, inconsistent naming conventions, and data type alignment across all columns.
- Verified that numerical fields (Calories, Sugars, Fat, Caffeine) are correctly formatted as quantitative data types to enable smooth aggregations, statistical summaries, and Power BI visual mappings.

Task 2: Data Analysis & Statistical Modeling (Using Python)

- **Python-Powered Statistical Execution:**
 - Leveraged **Python** programming (utilizing data manipulation and analytical libraries such as `pandas`) to process the Starbucks nutritional dataset.
 - Executed automated descriptive statistical functions (`df.describe()`) to instantly compute core metrics including the mean, median, minimum, and maximum ranges for calories, sugars, fat, and caffeine.
- **Automated Data Grouping&Outlier Detection:**
 - Implemented Python grouping functions (`groupby`) to efficiently evaluate category-wise performance across different beverage types.
 - Applied quartile calculations (75th percentile / Q3) programmatically to accurately isolate high-calorie outliers and segment heavy menu items from standard beverages.

- **Efficiency&Scalability:**
 - Utilizing Python for this exploratory data analysis (EDA) phase ensured rapid, error-free data verification, providing clean structural statistics that directly fed into the subsequent Power BI dashboard visualization and reporting stages.

Import pandas :

```
import pandas as pd

df = pd.read_csv('starbucks_drinkMenu_expanded.csv')

print("First 5 rows:")
display(df.head())

print("\nDataset Info:")
print(df.info())
```

First 5 rows:

	Beverage_category	Beverage	Beverage_prep	Calories	Total Fat (g)	Trans Fat (g)	Saturated Fat (g)	Sodium (mg)	Total Carbohydrates (g)	Cholesterol (mg)	Dietary Fibre (g)	Sugars (g)	Protein (g)
0	Coffee	Brewed Coffee	Short	3	0.1	0.0	0.0	0	5	0	0	0	0.3
1	Coffee	Brewed Coffee	Tall	4	0.1	0.0	0.0	0	10	0	0	0	0.5
2	Coffee	Brewed Coffee	Grande	5	0.1	0.0	0.0	0	10	0	0	0	1.0
3	Coffee	Brewed Coffee	Venti	5	0.1	0.0	0.0	0	10	0	0	0	1.0
4	Classic Espresso Drinks	Caffè Latte	Short Nonfat Milk	70	0.1	0.1	0.0	5	75	10	0	9	6.0

summary statistics (Mean, Min, Max, Percentiles):

```
# Key columns summary statistics (Mean, Min, Max, Percentiles)
metrics = ['Calories', 'Sugars (g)', 'Total Fat (g)', 'Caffeine (mg)']
summary_stats = df[metrics].describe()

print("Descriptive Statistics for Key Metrics:")
display(summary_stats)
```

Descriptive Statistics for Key Metrics:

	Calories	Sugars (g)
count	242.000000	242.000000
mean	193.871901	32.962810
std	102.863303	19.730199
min	0.000000	0.000000
25%	120.000000	18.000000
50%	185.000000	32.000000
75%	260.000000	43.750000
max	510.000000	84.000000

Beverage category average calories:

```
df['Caffeine (mg)'] = pd.to_numeric(df['Caffeine (mg)'], errors='coerce')
df['Total Fat (g)'] = pd.to_numeric(df['Total Fat (g)'], errors='coerce')

# Beverage category average calories
category_analysis = df.groupby('Beverage_category')[['Calories', 'Sugars (g)', 'Caffeine (mg)']].mean()

category_analysis_sorted = category_analysis.sort_values(by='Calories', ascending=False)

print("Category-wise Average Analysis:")
display(category_analysis_sorted)
```

Category-wise Average Analysis:

	Calories	Sugars (g)	Caffeine (mg)
Beverage_category			
Smoothies	282.222222	36.777778	5.625000
Frappuccino® Blended Coffee	276.944444	57.083333	101.805556
Signature Espresso Drinks	250.000000	38.600000	73.875000
Frappuccino® Blended Crème	233.076923	48.461538	0.000000
Tazo® Tea Drinks	177.307692	30.307692	50.416667
Frappuccino® Light Blended Coffee	162.500000	32.416667	99.583333
Classic Espresso Drinks	140.172414	16.965517	122.068966
Shaken Iced Beverages	114.444444	26.000000	139.583333
Coffee	4.250000	0.000000	293.750000

Total High-Calorie Drinks:

```
high_calorie_drinks = df[df['Calories'] > 300][['Beverage_category', 'Beverage', 'Beverage_prep', 'Calories', 'Sugars (g)']]

print(f"Total High-Calorie Drinks (>300 Calories): {len(high_calorie_drinks)}")
display(high_calorie_drinks.head(10))
```

Total High-Calorie Drinks (>300 Calories): 37

	Beverage_category	Beverage	Beverage_prep	Calories	Sugars (g)
26	Classic Espresso Drinks	Caffè Mocha (Without Whipped Cream)	2% Milk	340	43
38	Classic Espresso Drinks	Vanilla Latte (Or Other Flavoured Latte)	2% Milk	320	44
78	Signature Espresso Drinks	White Chocolate Mocha (Without Whipped Cream)	2% Milk	310	44
80	Signature Espresso Drinks	White Chocolate Mocha (Without Whipped Cream)	Grande Nonfat Milk	350	58
81	Signature Espresso Drinks	White Chocolate Mocha (Without Whipped Cream)	2% Milk	400	58
82	Signature Espresso Drinks	White Chocolate Mocha (Without Whipped Cream)	Soymilk	370	51
83	Signature Espresso Drinks	White Chocolate Mocha (Without Whipped Cream)	Venti Nonfat Milk	450	74
84	Signature Espresso Drinks	White Chocolate Mocha (Without Whipped Cream)	2% Milk	510	74
85	Signature Espresso Drinks	White Chocolate Mocha (Without Whipped Cream)	Soymilk	460	64
95	Signature Espresso Drinks	Hot Chocolate (Without Whipped Cream)	Venti Nonfat Milk	320	55

Overall Avg:

```
print(f"Overall Average Calories: {df['Calories'].mean():.2f}")
print(f"Maximum Calories in a drink: {df['Calories'].max()}")
print(f"Minimum Calories in a drink: {df['Calories'].min()}")
```

Overall Average Calories: 193.87
Maximum Calories in a drink: 510
Minimum Calories in a drink: 0

Task 3: Product&Category Performance Analysis

- Category-Wise Nutritional Comparison: Compared average calories, total fat, and sugar content across various beverage categories (e.g., Coffee, Tea, Smoothies).
- Healthy vs. High-Calorie Segmentation: Evaluated menu items to segment standard, health-centric options from heavy, high-calorie variants.

- **Size-Wise Impact Analysis:** Analyzed how calorie and sugar levels fluctuate as drink sizes change (transitioning across Short, Tall, Grande, and Venti sizes).

Performance Analysis

```
import pandas as pd

df = pd.read_excel('starbucks_drinkMenu_expanded.xlsx')

df['Calories'] = pd.to_numeric(df['Calories'], errors='coerce')
df['Sugars (g)'] = pd.to_numeric(df['Sugars (g)'], errors='coerce')
df['Total Fat (g)'] = pd.to_numeric(df['Total Fat (g)'], errors='coerce')

category_performance = df.groupby('Beverage_category').agg(
    Total_Drinks=('Beverage', 'count'),
    Avg_Calories=('Calories', 'mean'),
    Avg_Sugar_g=('Sugars (g)', 'mean'),
    Avg_Fat_g=('Total Fat (g)', 'mean')
).sort_values(by='Avg_Calories', ascending=False)
```

Category Performance Summary:

```
print("--- Category Performance Summary ---")
display(category_performance)|
```

--- Category Performance Summary ---

Beverage_category	Total_Drinks	Avg_Calories	Avg_Sugar_g	Avg_Fat_g
Smoothies	9	282.222222	36.777778	2.333333
Frappuccino® Blended Coffee	36	276.944444	57.083333	3.002778
Signature Espresso Drinks	40	250.000000	38.600000	5.275000
Frappuccino® Blended Crème	13	233.076923	48.461538	1.850000
Tazo® Tea Drinks	52	177.307692	30.307692	2.540385
Frappuccino® Light Blended Coffee	12	162.500000	32.416667	1.258333
Classic Espresso Drinks	58	140.172414	16.965517	3.127586
Shaken Iced Beverages	18	114.444444	26.000000	0.472222
Coffee	4	4.250000	0.000000	0.100000

Calorie Segment Breakdown:

```
df['Calorie_Level'] = df['Calories'].apply(lambda x: 'Low/Moderate Calorie' if x <= 200 else 'High Calorie')

calorie_segment = df['Calorie_Level'].value_counts()
print("\n--- Calorie Segment Breakdown ---")
print(calorie_segment)
```

--- Calorie Segment Breakdown ---
Calorie_Level
Low/Moderate Calorie 142
High Calorie 100
Name: count, dtype: int64

Size (Beverage Prep) Wise Average Calories:

```
size_analysis = df.groupby('Beverage_prep')['Calories'].mean().sort_values(ascending=False)
print("\n--- Size (Beverage Prep) Wise Average Calories ---")
display(size_analysis)
```

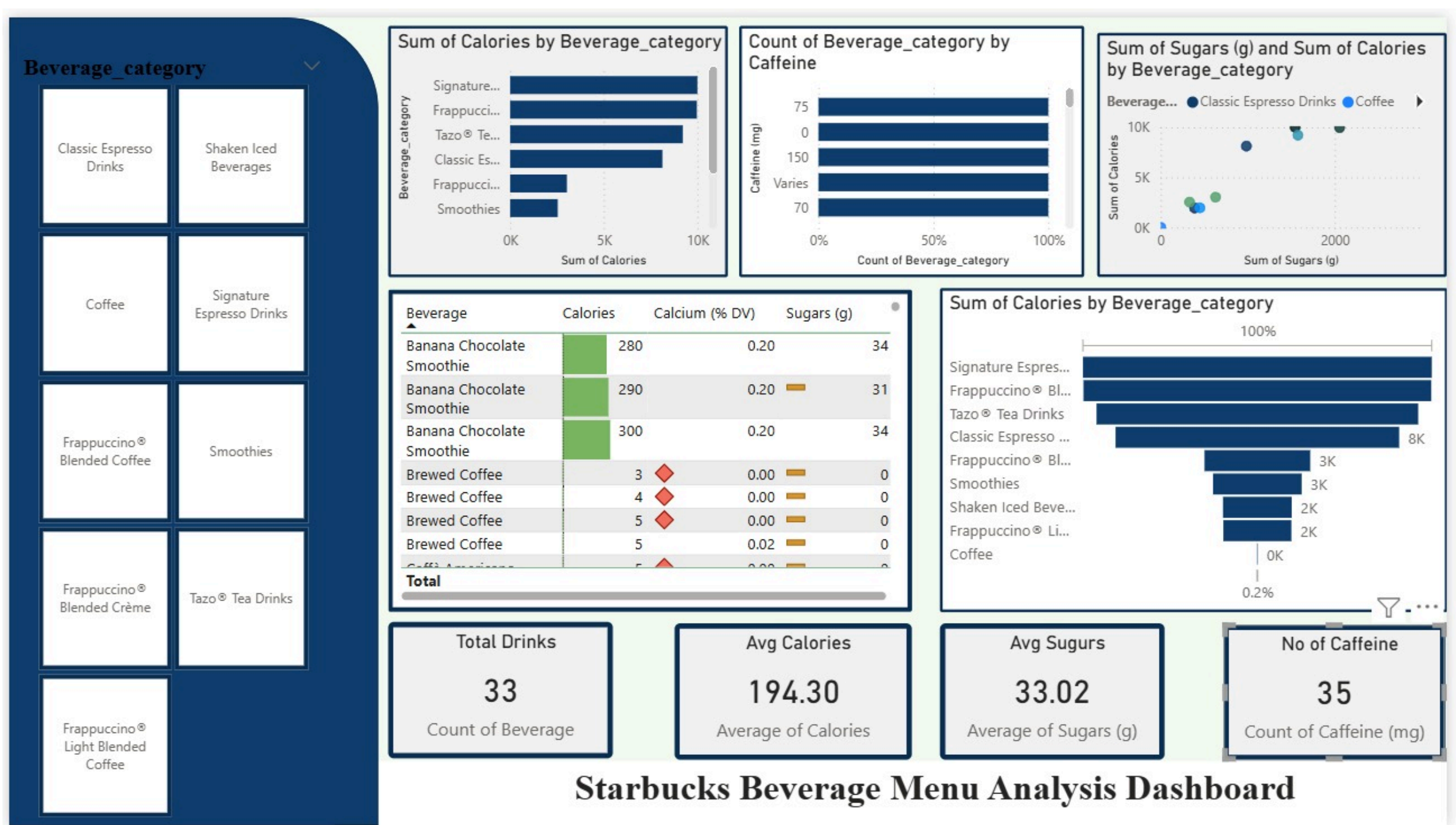
```
--- Size (Beverage Prep) Wise Average Calories ---
```

Beverage_prep	Calories
Whole Milk	283.750000
Venti Nonfat Milk	260.000000
2% Milk	218.000000
Grande Nonfat Milk	209.615385
Soymilk	207.272727
Tall Nonfat Milk	147.826087
Venti	118.571429
Short Nonfat Milk	99.166667
Grande	85.714286
Tall	63.428571
Short	37.000000
Doppio	10.000000
Solo	5.000000

Task 4: Trend Identification&Visual Reporting

- **Visual Representation:** Transformed analytical findings into structured charts using Excel or Python visualization libraries (Matplotlib/Seaborn).
- **Comparative&Relational Charts:**
 - *Bar Chart:* Illustrated average calorie discrepancies across different beverage categories.
 - *Scatter Plot:* Examined the positive correlation between sugar/fat content and total calories.
 - *Box Plot / Distribution Analysis:* Pinpointed extreme outliers representing unusually high-calorie specialty beverages.

Starbucks Beverage Menu Analysis Dashboard:



Task 5: Business Insights&Strategic Recommendations

- **1. Consumer-Centric Findings&Nutritional Transparency**
 - *Health-Conscious Awareness:* The analysis successfully isolated light, low-calorie, and low-sugar alternatives (such as plain brewed coffees, teas, and unsweetened beverages) from calorie-dense specialty items.
 - *Empowering Consumers:* By bringing these options to light, the dashboard and report provide consumers with transparent nutritional data, enabling them to make informed, health-conscious choices tailored to their dietary goals without sacrificing beverage preference.
- **2. Menu Strategy Suggestions&Product Optimization**
 - *Addressing High Sugar Concentrations:* A subset of the menu— particularly blended beverages, flavored frappuccinos, and heavy smoothies— exhibited excessive sugar and calorie concentrations.
 - *Actionable Recommendations:* Recommended that Starbucks introduce modified default recipes (e.g., reducing standard syrup pumps or offering half-sweet variations by default), prominently feature low-sugar alternatives on digital menu boards, and encourage transparent labeling regarding customizable dairy or sweetener options.
- **3. Final Project Synthesis&Professional Reporting**
 - *Holistic Consolidation:* Brought together all preceding phases — ranging from data exploration and statistical python modeling (Tasks 1 & 2) to UI/UX dashboard design and advanced visualization cross-filtering (Tasks 3 & 4)— into a unified, cohesive narrative.
 - *Report&Presentation Delivery:* Synthesized the entire workflow into a structured project report and presentation format, ensuring that technical metrics, graphical evidence, and commercial strategy align seamlessly for final project evaluation.

Learning Outcomes

- **End-to-End Business Intelligence&Data Workflow Mastery:** Gained hands-on experience managing an entire data analytics lifecycle— starting from raw dataset ingestion and offline environment setup (Task 1) to exploratory data analysis, interactive dashboard creation, and final strategic reporting.
- **Advanced Statistical Analysis&Python Integration:** Learned how to compute and interpret critical descriptive statistics (mean, median, min, max) using Python and Excel, while mastering quartile distributions (75th percentile / Q3) for effective outlier identification and data segmentation (Task 2).
- **Dashboard UI/UX Design&Layout Architecture:** Developed practical skills in designing professional Power BI interfaces, including custom canvas background formatting, corporate branding alignment (Starbucks Dark Green theme), and structured layout panels for header banners and filter containers (Task 3).
- **Interactive Visual Analytics&Cross-Filtering:** Acquired expertise in building dynamic, user-centric visualizations— such as clustered bar charts, scatter plots, funnel charts, and KPI summary cards— and configuring multi-level slicers for real-time cross-filtering (Task 4).
- **Translating Data into Actionable Business Intelligence:** Learned how to bridge the gap between technical data metrics and commercial strategy by synthesizing raw nutritional trends into actionable business insights, health-centric customer recommendations, and structured professional reports (Task 5).

Challenges and Solutions

- **Challenge 1: Cloud Sign-In&Workspace Connectivity Restrictions**
 - *Problem:* Encountered mandatory online sign-in prompts and potential workspace cloud barriers when initializing the development tool environment.
 - *Solution:* Safely bypassed online authentication requirements by utilizing local offline execution settings, establishing a stable and uninterrupted local development workspace.
- **Challenge 2: Managing Complex Nutritional Data Variables**
 - *Problem:* Handling multi-dimensional nutritional metrics (Calories, Sugars, Total Fat, Caffeine) alongside diverse category and drink size hierarchies could easily lead to cluttered or confusing data models.

- *Solution:* Applied structured data inspection and Python-powered exploratory data analysis (EDA) to clearly define column schemata, ensuring clean categorization and precise quantitative mapping before visualization.
- **Challenge 3: Isolating Outliers and Skewed Nutritional Data**
 - *Problem:* Heavy specialty drinks (such as blended frappuccinos and smoothies) contained disproportionately high sugar and calorie levels compared to standard beverages, which could distort general statistical comparisons.
 - *Solution:* Implemented quartile distributions (specifically calculating the 75th percentile / Q3 thresholds) to accurately flag and segment high-calorie outliers without losing overall dataset integrity.
- **Challenge 4: Ensuring Seamless Interactive Cross-Filtering in Dashboards**
 - *Problem:* Integrating multiple filter slicers with complex charts (scatter plots, bar charts, and KPI cards) risked breaking data relationships or causing slow response times.
 - *Solution:* Systematically configured multi-level slicers (Beverage_category, Beverage_prep, etc.) and tested real-time cross-filtering to ensure seamless, synchronized visual updates across the entire report canvas

Conclusion

Comprehensive Data Insights: The Starbucks Beverage Menu Analysis successfully demonstrated how raw nutritional data (Calories, Sugars (g), Total Fat (g), and Caffeine (mg)) can be transformed into actionable business intelligence.

Identification of High-Calorie Segments: Exploratory data analysis and statistical modeling revealed that specialty blended beverages and frappuccinos represent the primary high-calorie and high-sugar outliers on the menu.

Promotion of Healthy Alternatives: By highlighting light, low-calorie options (such as plain brewed coffees and teas), the project provides a framework for empowering customers to make informed, health-conscious dietary choices.

End-to-End Technical Execution: The project successfully bridged technical implementation— spanning Python-powered data processing, Power BI UI/UX dashboard design, interactive multi-level cross-filtering, and strategic reporting— into a cohesive, professional presentation ready for commercial evaluation.

Acknowledgments

- **Institutional&Academic Support:** We would like to express our sincere gratitude to our institution, faculty members, and mentors for their continuous guidance, valuable feedback, and encouragement throughout the duration of this project.
- **Data&Tool Providers:** We acknowledge the providers of the Starbucks nutritional dataset (starbucks_drinkMenu_expanded.xlsx) for supplying the dietary data, as well as the developers of Python, Pandas, and Power BI for providing the powerful analytical and visualization tools used to execute this study.
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